

Day 4 - Mini Project Documentation

Introduction

This day focused on building a complete console-based Student Progress Tracker in C. The project was extended step by step through multiple tasks, improving functionality, structure, and logic.

Task 1 - Multiple Records

Implemented storage of multiple students using an array with a fixed capacity. Added functionality to insert multiple students and display all stored records. Maintained a count variable to track current number of students.

Task 2 - Enum and Validation

Introduced enum for student status (Not Started, In Progress, Completed). Improved input validation for menu and user input. Used switch statements for cleaner control flow.

Task 3 - Data Analysis

Added report functionality using loops and conditions. Calculated total students, completed students, average progress, maximum and minimum values. Classified performance based on average.

Task 4 - Pointer Usage

Implemented pointer-based update function. Allowed modification of student data (progress and status) using references instead of copies. Ensured changes persist after function execution.

Task 5 - Search and Evaluation

Added search functionality by ID. Displayed specific student data. Implemented evaluation logic using combined conditions to generate feedback messages.

Task 6 - Sorting and Ranking

Implemented sorting of students based on progress using nested loops. Added ranking display showing ordered students from highest to lowest progress. Ensured compatibility with existing features.

Task 7 - Edit and Delete

Extended project to allow full data management. Implemented deletion of a student by ID. Shifted array elements after deletion to maintain consistency. Updated count variable correctly. Ensured all features continue to work after deletion.

Conclusion

The project evolved into a complete and structured system. It supports adding, updating, searching, analyzing, sorting, and deleting student data. All features are integrated into a single menu-driven application.